

Task 3 and Task 4

Task 3 focused on upgrading the basic sensor prototype into a **reliable and robust smart automation system** at the local device level. The primary goal was to eliminate the instability inherent in the raw PIR sensor output, which often led to momentary alarm resets ("chatter"). This was solved by implementing a custom **2.5-second Co-down Timer (kPIR_RESET_TIME_MS)** in the firmware. This essential automation logic forces the alarm outputs (the four blinking LEDs and the tone buzzer) to remain active for a minimum duration after the PIR signal drops, ensuring the alarm does not prematurely shut off. Furthermore, the task ensured all output functions utilized **non-blocking millis() timing** for the LED blinking and the tone() function for the buzzer, guaranteeing that the core alarm logic remains fast and uninterrupted by the output mechanisms.

Task 4 completed the project by transforming the stable Task 3 automation system into a fully **cloud-connected Internet of Things (IoT) device**. This was achieved by integrating the ESP32 with the **Blynk IoT Platform**. The core achievement involved mapping the local system's state to cloud-accessible data streams using Virtual Pins. The alarm's binary status was sent to the cloud via **Virtual Pin V0** (used by the dashboard's LED widget), while all event messages (alarm trigger, reset) were transmitted via **Virtual Pin V1** (used by the Terminal widget). This integration provided essential cloud-based features such as **real-time monitoring** of the alarm status from anywhere in the world and the ability to set up **push notifications** based on the V0 state, successfully fulfilling the objective of creating a remote monitoring solution.